

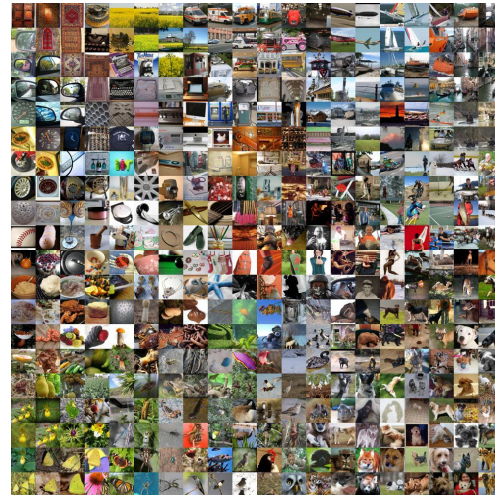
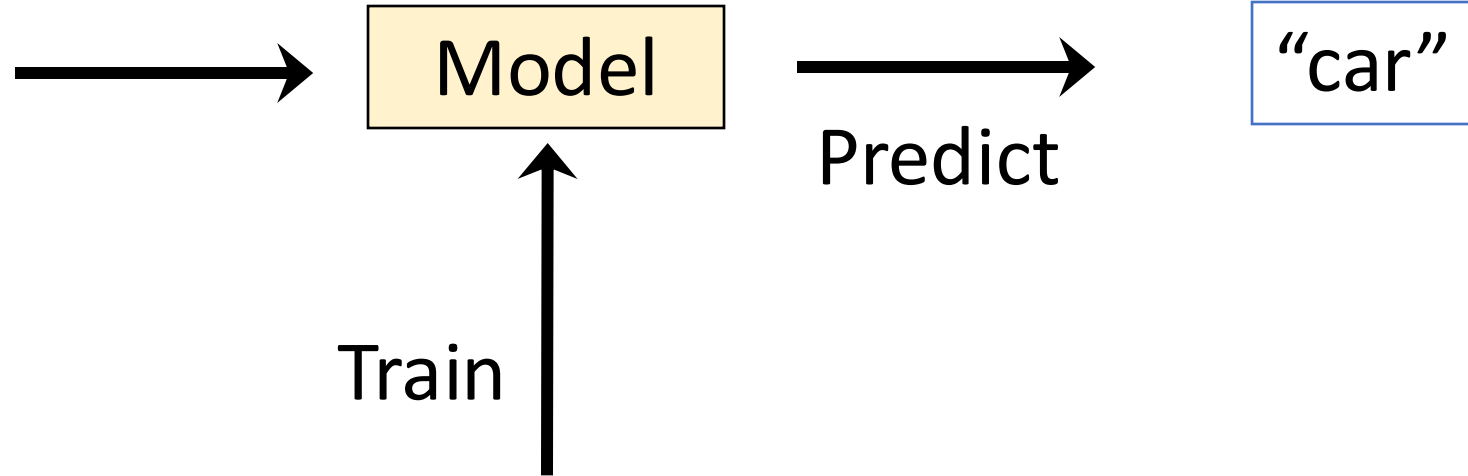
Modeling & Computation

Erisa Terolli

Learning from Data



Future Data



Past Data
(perhaps with **labels**)

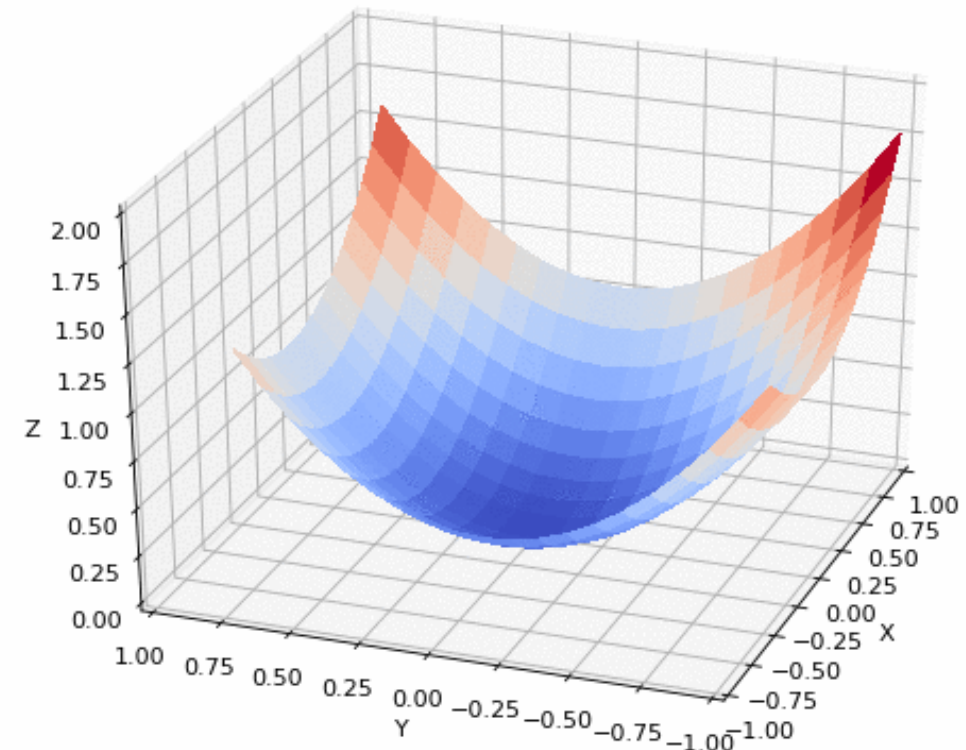
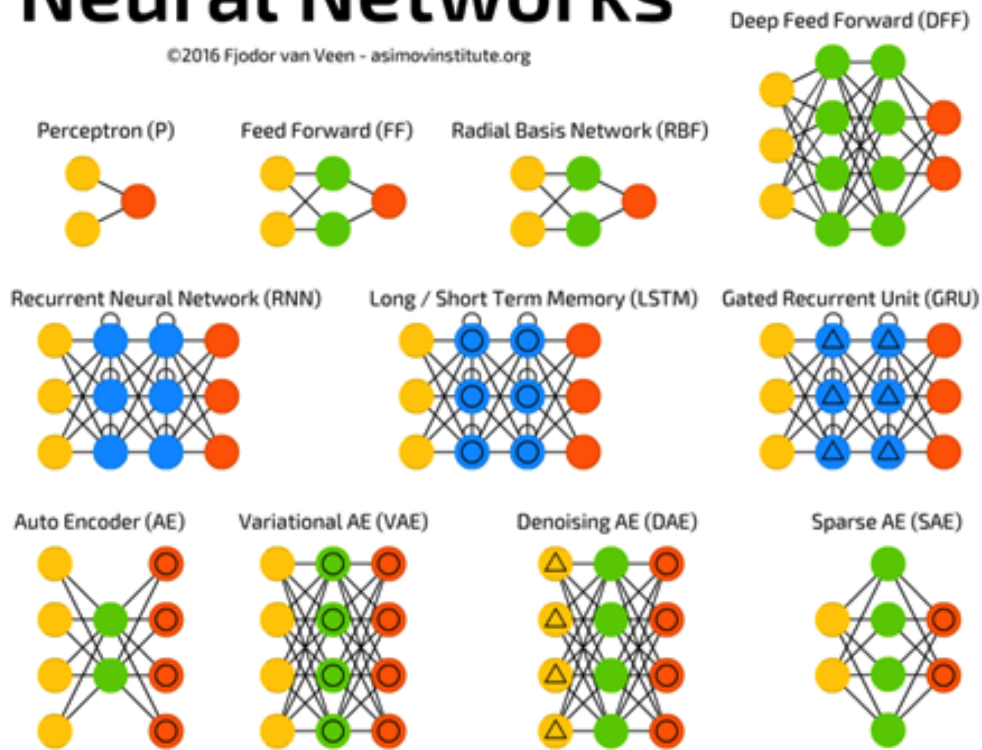
Machine Learning in Practice

Modeling

Computation

Neural Networks

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Modeling

Linear Models

- Feature vector: $\mathbf{x} \in \mathbb{R}^d$ (e.g., features of a house).

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix}$$

of bedrooms
of bathrooms
square feet
$\vdots$
age of house

Linear Models

- Feature vector: $\mathbf{x} \in \mathbb{R}^d$ (e.g., features of a house).
- Prediction: $f(\mathbf{x}) = \mathbf{x}^T \mathbf{w}$ (e.g., housing price).

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix} \begin{array}{l} \text{\# of bedrooms} \\ \text{\# of bathrooms} \\ \text{square feet} \\ \vdots \\ \text{age of house} \end{array} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_d \end{bmatrix}$$

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- $f(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \dots + w_d x_d$
- $w_1, w_2, \dots, w_d$: weights

Linear Models

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Question: How to find $\mathbf{w}$?



Price = \$0.5M

Features of a House

$$\mathbf{x} \in \mathbb{R}^d$$

Prediction:

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{w}$$

Linear Models

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Question: How to find $\mathbf{w}$?

- Training features: $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.
- Training targets: $y_1, \dots, y_n \in \mathbb{R}$.



...



totally n houses

Linear Models

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- Loss function: $L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} - y_i)^2$.

Linear Models

- Feature vector: $\mathbf{x} \in \mathbb{R}^d$ (e.g., features of a house).
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- Loss function: $L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} - y_i)^2$.
- Least squares regression: $\mathbf{w}^* = \min_{\mathbf{w}} L(\mathbf{w})$.

Beyond Linear Models

Linear Models Are Not Expressive

Example: Given a person's photo, predict her/his age.



Photo (features)



Age = 36

Linear Models Are Not Expressive

Example: Given a person's photo, predict her/his age.



Age = 36

Photo (features)

Question: Can we use linear regression?

- Linear regression assumes the **target** is a weighted average of **every pixel in the photo**.
- Linear regression works poorly for age prediction.

Traditional Approaches



SIFT Feature



HOG Feature



LBP Feature



Traditional Approaches



SIFT Feature



HOG Feature



LBP Feature

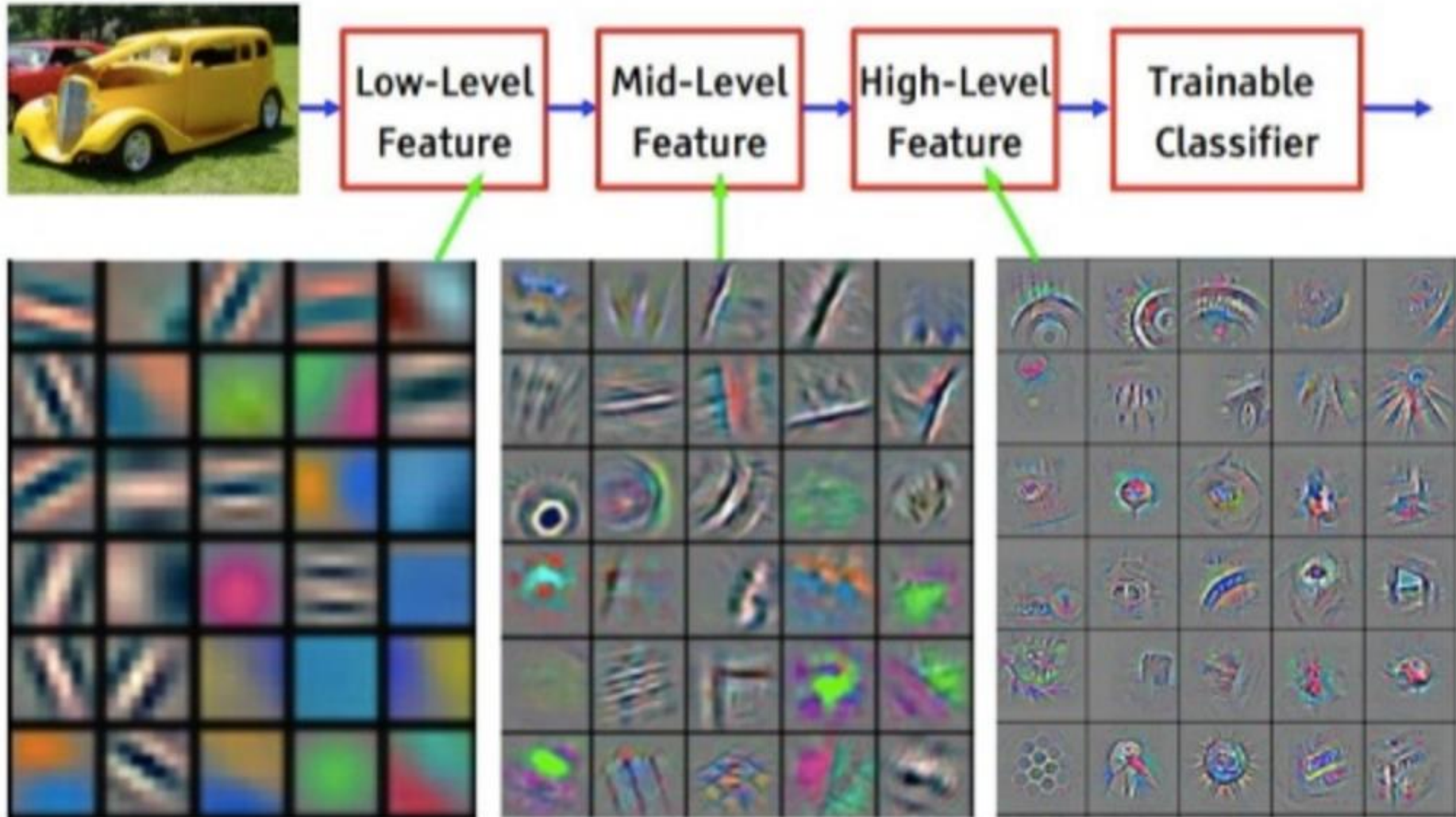


Linear Model



“dog”

Convolutional Neural Networks (CNNs)

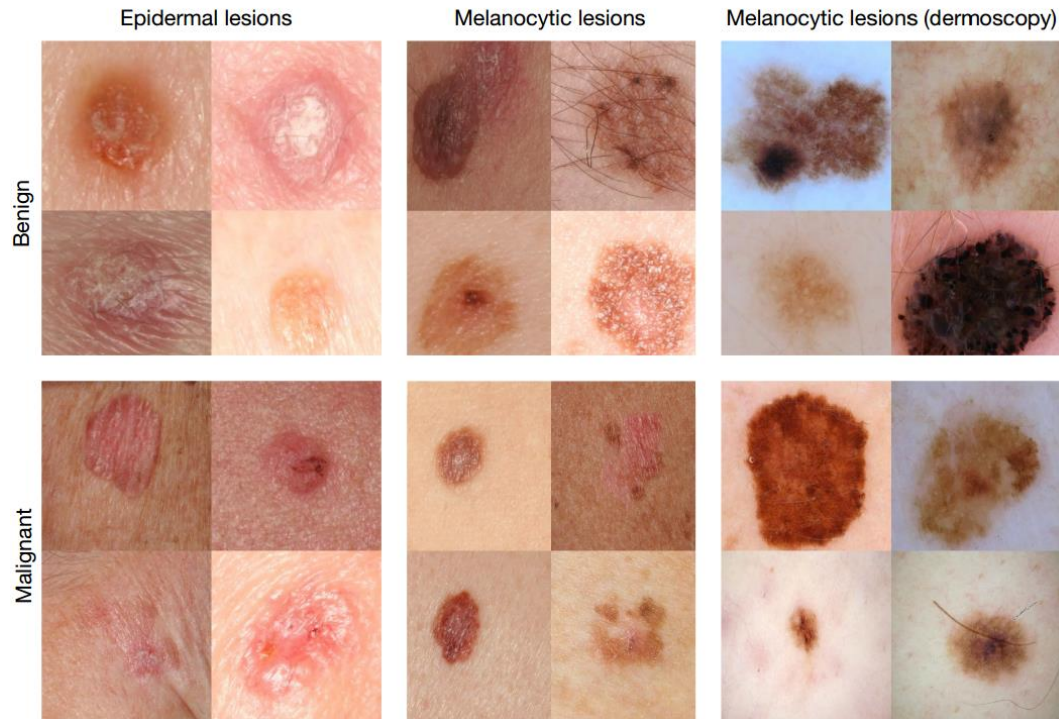


Applications of CNNs



- CNNs are suitable for image data.
- CNNs convert images to effective representations. (Feature extraction.)
- Applications:
 - Image/video recognition.
 - Face recognition.
 - Image generation.
 - ...

Applications of CNNs: Medical Diagnosis



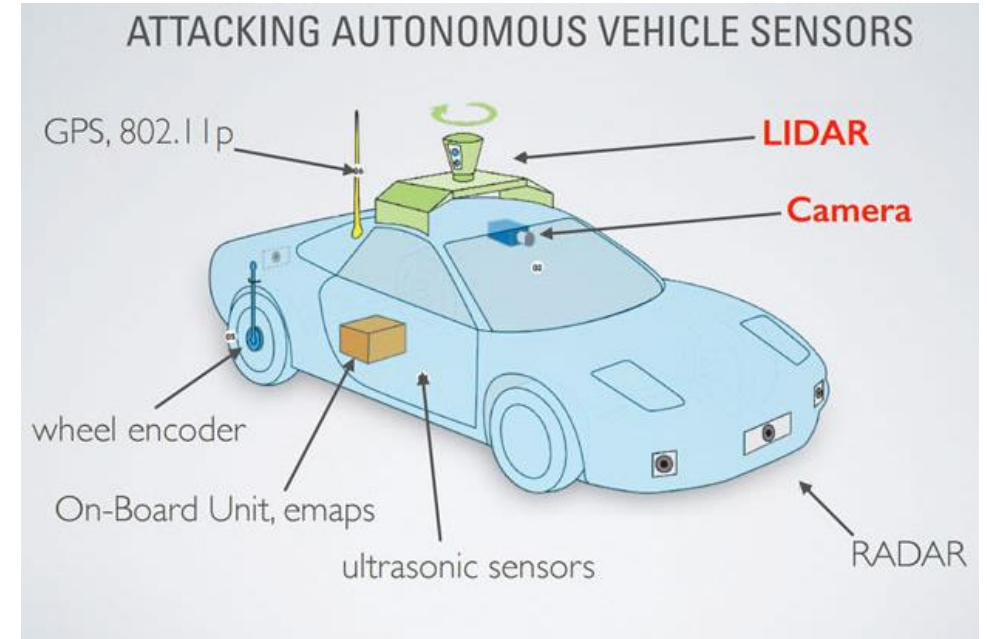
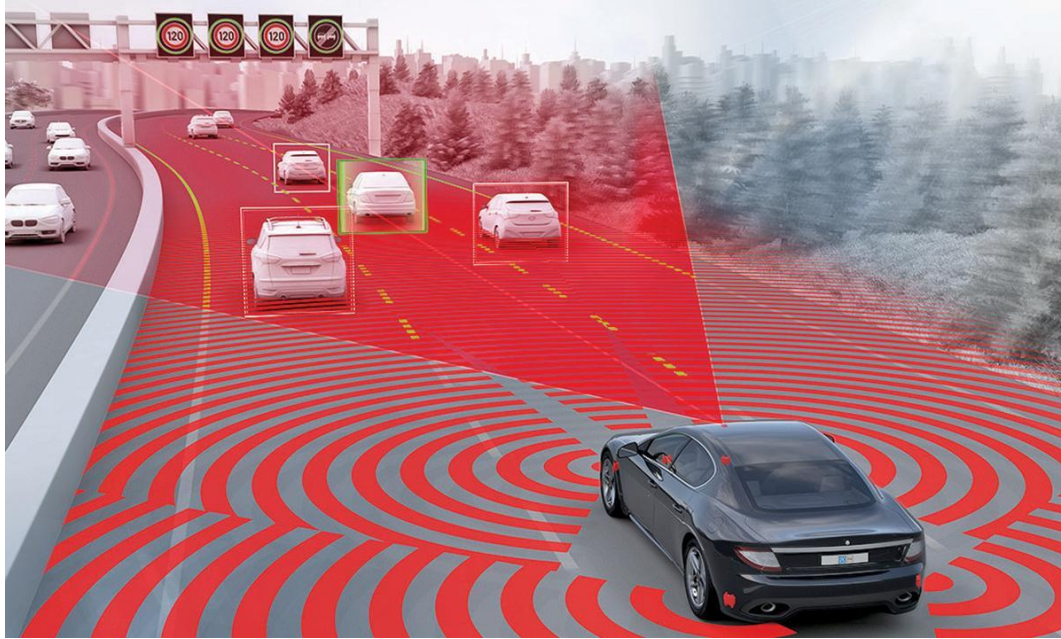
Example: Skin cancer diagnosis

- Input: an image.
- Outputs:
 - Is it skin cancer?
 - Benign or malignant?
- The same accuracy as human experts.

Reference

[1] Esteva et al. [Dermatologist-level classification of skin cancer with deep neural networks](#). *Nature*, 2017.

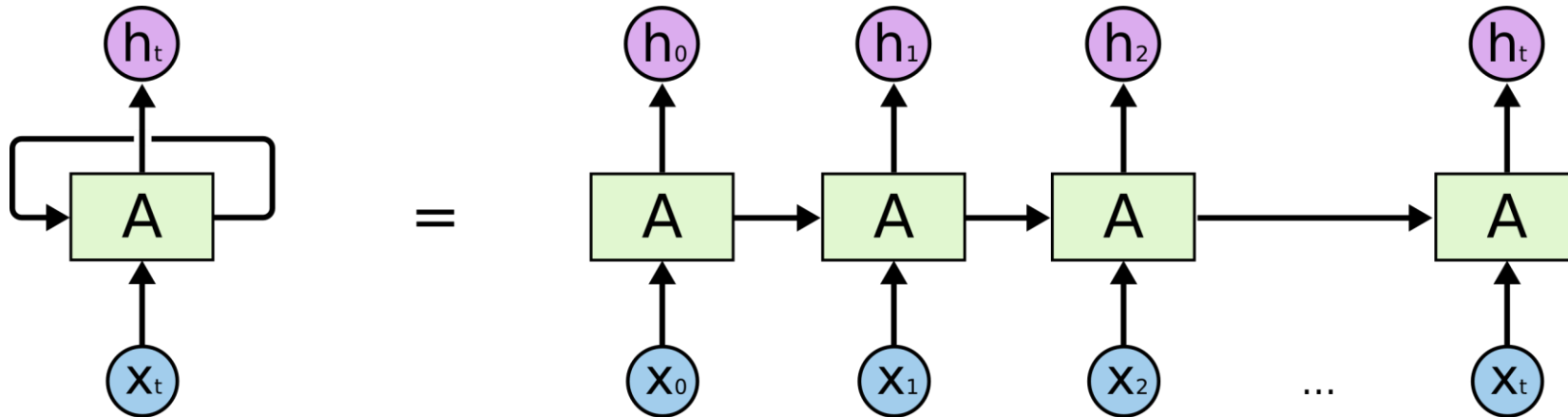
Applications of CNNs: Self Driving Cars



- CNNs play an important role in self-driving cars.
 - Understand images taken by the cameras.
 - Recognize signs, cars, pedestrians, and obstacles.
- CNN is not everything; self-driving car is a sophisticated system.

Recurrent Neural Networks (RNNs)

- RNNs naturally fit sequence data, e.g.,
 - time series data (e.g., stock price, weather),
 - text data,
 - speech data...



Applications of RNNs: Machine Translation



Applications of RNNs: Machine Translation



Applications of RNNs: Speech Recognition



amazon alexa



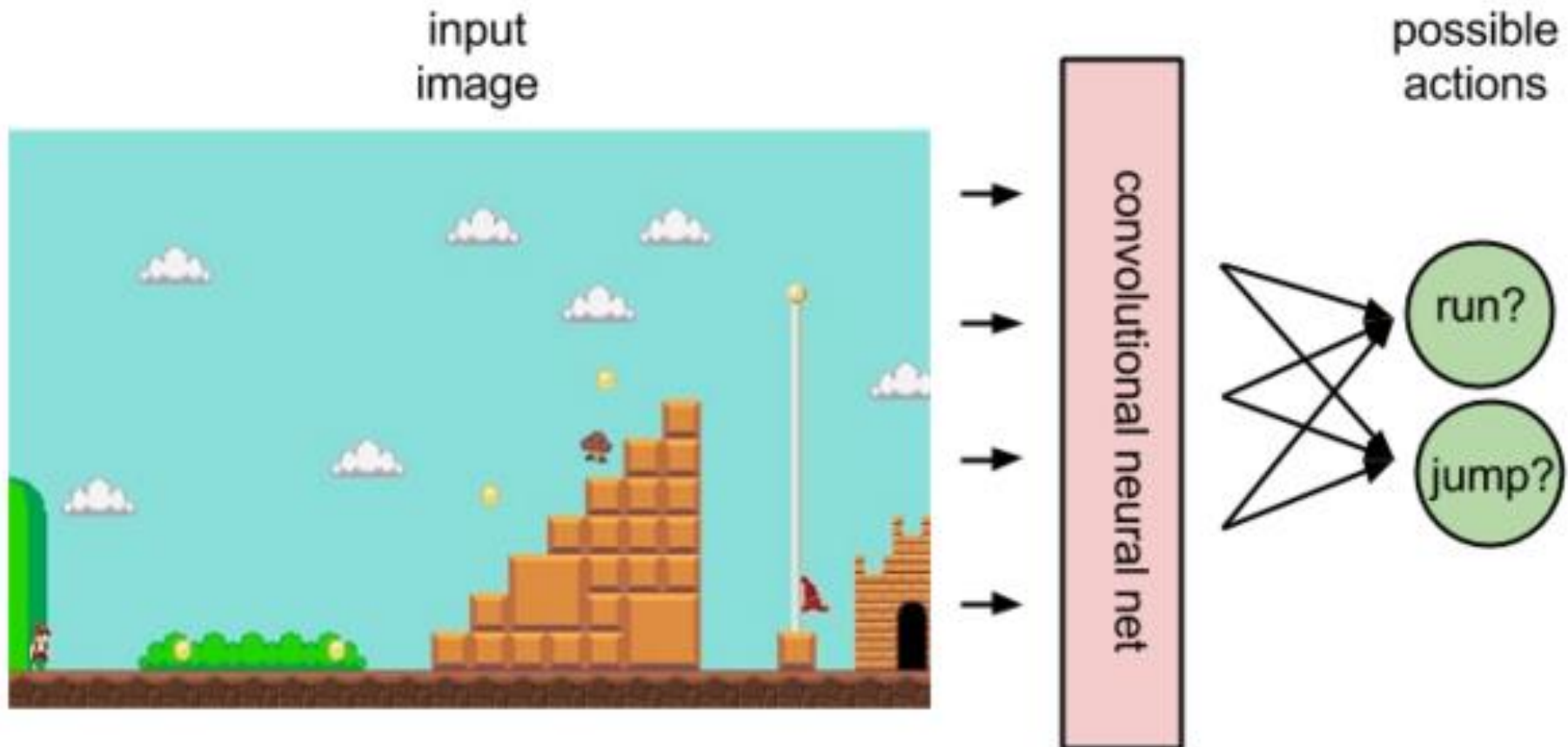
Hi, I'm Cortana.



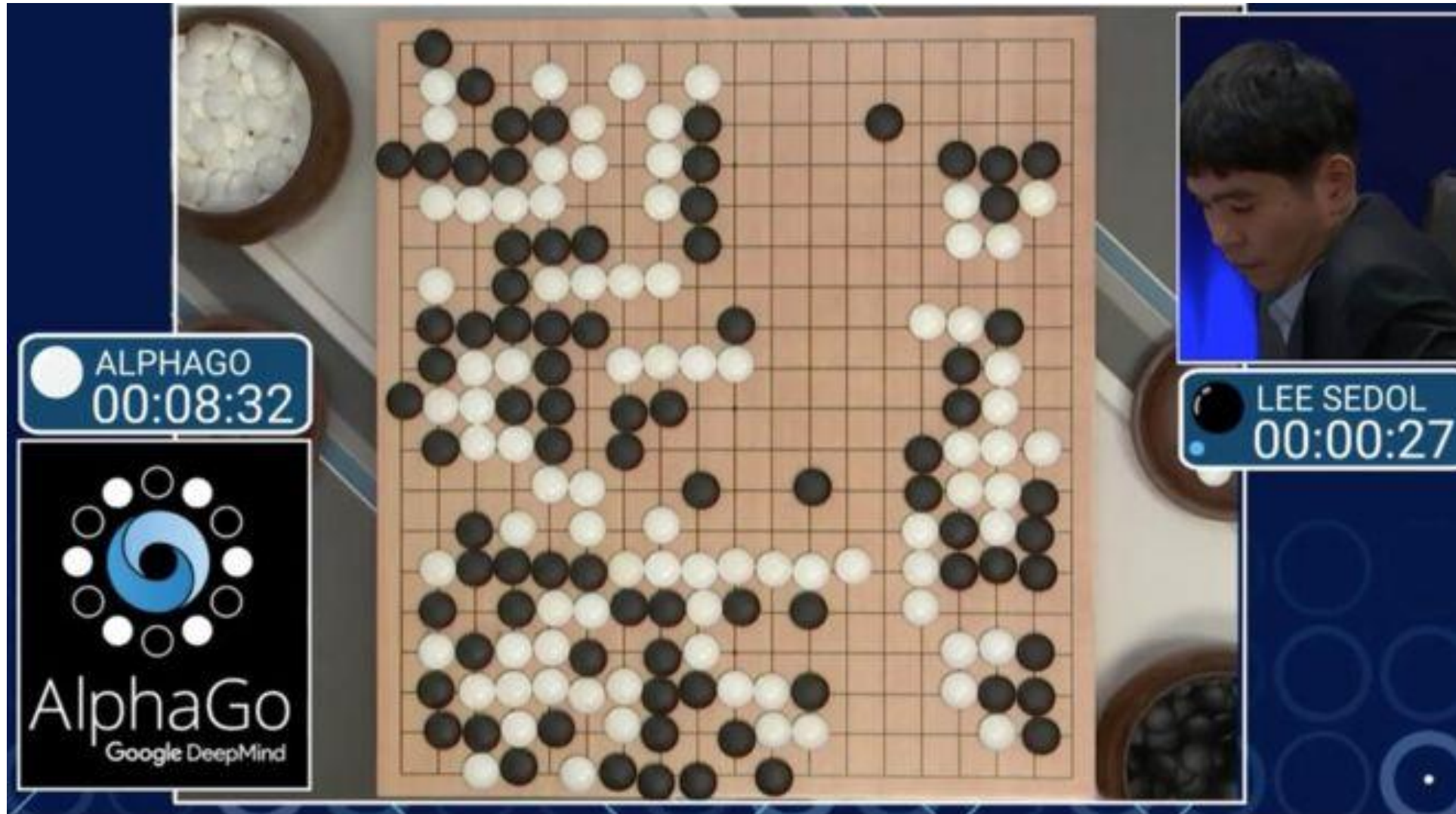
Hey Siri

Deep Reinforcement Learning (DFL)

- DFL has applications in robotics, video game, and finance.



Applications of DRL: Games



What we have learned so far...

- **ML tasks**: regression, classification, ...
- **ML models**: linear models, CNNs, RNNs, ...

Example: least squares regression model

- Loss function: $L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} - y_i)^2$.
- Optimization: $\mathbf{w}^* = \min_{\mathbf{w}} L(\mathbf{w})$.

- How to solve the model?

Computations

Computational Methods

- **ML tasks**: regression, classification, ...
- **ML models**: linear models, CNNs, RNNs, ...

Example: least squares regression model

- Loss function: $L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} - y_i)^2$.
- Optimization: $\mathbf{w}^* = \min_{\mathbf{w}} L(\mathbf{w})$.

- **Computations**: solve the model using numerical algorithms, e.g., gradient descent (GD) or stochastic descent (SGD).

Computational Challenges

- **Big data**: too many training samples.
 - ImageNet: **14 million** 256×256 images.
- **Big model**: too many model parameters.
 - ResNet-50 (a very popular CNN architecture) has **25 million parameters**.

Computational Challenges

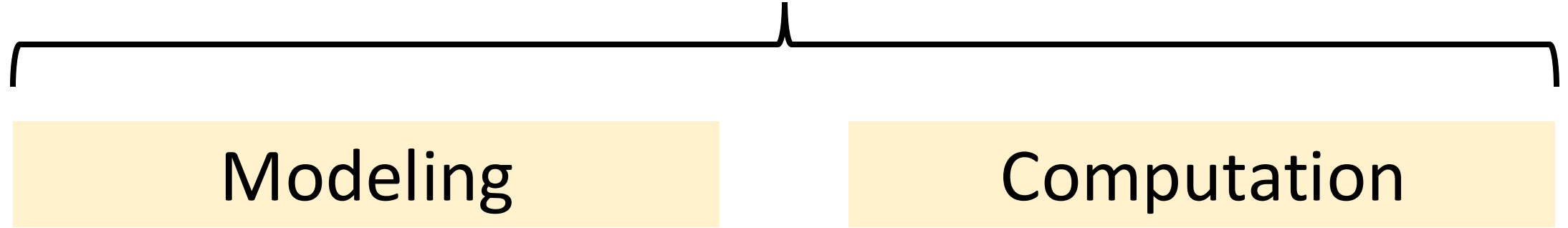
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- Training ResNet-50 on ImageNet using a **single GPU** takes around **14 days**.

Computational Challenges

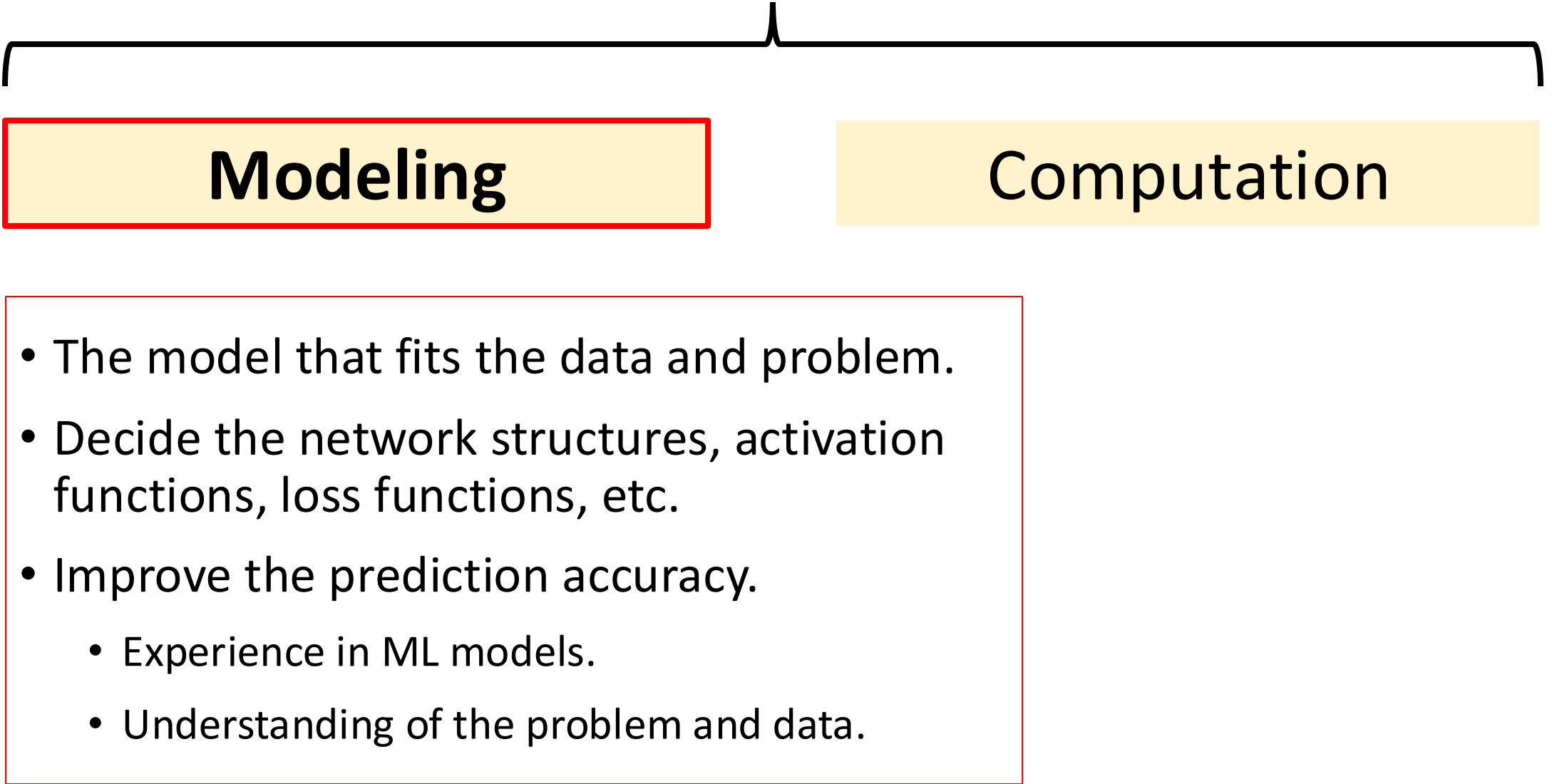
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➔ Efficient algorithms and software systems are necessary.

Machine Learning in Practice



Machine Learning in Practice

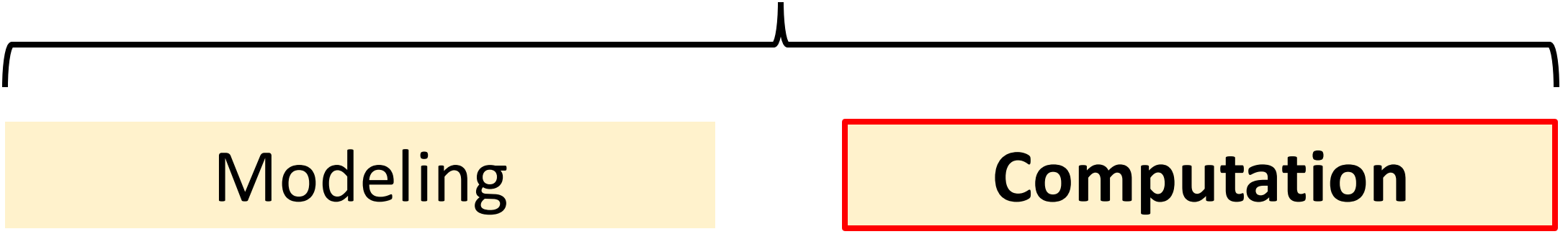


Modeling

Computation

- The model that fits the data and problem.
- Decide the network structures, activation functions, loss functions, etc.
- Improve the prediction accuracy.
 - Experience in ML models.
 - Understanding of the problem and data.

Machine Learning in Practice



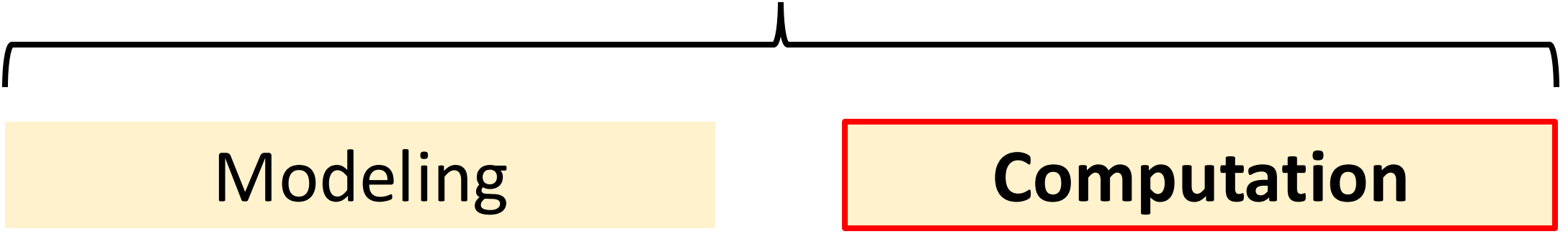
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graph TD; ML[Machine Learning in Practice] --- Modeling[Modeling]; ML --- Computation[Computation]; style Computation stroke:#f00,stroke-width:2px
```

Modeling

Computation

- Design or apply efficient algorithms.
- Implement the algorithm using systems like TensorFlow, Hadoop, etc.

Machine Learning in Practice



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graph TD; ML[Machine Learning in Practice] --- Modeling[Modeling]; ML --- Computation[Computation]; style Computation stroke:#f00,stroke-width:2px
```

Modeling

Computation

- Design or apply efficient algorithms.
- Implement the algorithm using systems like TensorFlow, Hadoop, etc.
- Optimize your code.
 - Experience in algorithms.
 - Experience in systems.